Ghost Seal

AI-disclosure standard for generated imagery — process, specification and verification — v1.0, 2026-08-05

1. Purpose

Every image we publish that was made with a generative model must disclose that fact. The disclosure has to satisfy two audiences that pull in opposite directions:

- **A regulator or complainant** must be able to find an unambiguous disclosure on the asset itself, and a machine must be able to read it without human help.
- **A customer** must not be pushed away by a warning label stuck on our marketing.

The Ghost Seal resolves this by separating the two jobs. The **machine-readable metadata** carries the legal weight: an explicit, standards-based declaration that the image is synthetic. The **visible seal** carries the human-facing courtesy: a small carved stamp in the corner, findable on inspection, ignorable at a glance.

The visible mark is deliberately quiet. The provenance record is deliberately loud.

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2. Regulatory context, stated honestly

The EU AI Act's transparency obligations (Article 50) work on two levels: synthetic image content should be **marked in a machine-readable format** so it is detectable as artificially generated, and certain publications must additionally be **disclosed to the viewer**. The IPTC `digitalSourceType` vocabulary term `trainedAlgorithmicMedia` is the interoperable industry expression of the machine-readable part, and it is what we embed.

What this document does **not** claim:

- It does not certify us as compliant. It implements marking practice that is consistent with the regulation's intent and with industry standards.
- It does not decide whether a given asset triggers Article 50 duties at all, or whether we act as provider or deployer for that asset. That is a question for counsel, and the answer may differ per campaign.
- The metadata we write is **unsigned**, therefore strippable. If we need a tamper-evident chain, C2PA signing with a company certificate goes on top of this; it does not replace it.

Treat this as the engineering half of compliance. The legal half needs sign-off.

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3. How we got to the Ghost Seal

The seal was not the first idea. Six disclosure treatments were built at full resolution on a real campaign asset (a 1792x2400 product ad), then measured rather than debated. Two numbers drove the decision:

• **∆L* on ink** — the mean CIE L* (perceptual lightness) difference between the mark and the background it sits on. This is how much the mark "pops".

• **Thumbnail RMS** — the root-mean-square pixel difference between the marked and unmarked frame after downscaling to 1/10 scale. This approximates how much of the mark survives into a feed-sized thumbnail, i.e. how much it costs us in the scroll.

Candidate	∆L* on ink	Thumbnail RMS	Verdict
Glass etch	8.6	0.45	Very quiet, but reads as a plain text label
Frosted micro-badge	4.6	0.72	Clean, but a UI chip on a print-grade ad
Folded corner ribbon	4.5	0.84	Rejected: looks like a stock-photo sample watermark
Hologram seal	7.7	1.57	Chosen direction: reads as a certification, not a warning
Edge caption	14.9	1.11	Explicit, but obvious legal fine print
Legal strip below frame	n/a	0.00	Strongest compliance, changes the composition

The hologram seal won on positioning: a seal reads as *provenance* and even as a quality mark, where a caption reads as *disclaimer*. But at first build it was too loud — 8.3% of frame width with a saturated iridescent disc.

Four progressively quieter revisions followed, each dropping the "sticker" qualities: the solid fill went away, the rim went away, the iridescence was reduced to a single diagonal sheen band, and the seal geometry was re-expressed as **pure luminance relief** — the mark became something pressed into the surface rather than placed on top of it.

Revision	Diameter	∆L*	Thumbnail RMS
Original holo seal	149px (8.3% W)	7.7	1.57
D — visible	122px (6.8% W)	4.4	0.66
C — foil	108px (6.0% W)	3.4	0.42
B — carved	99px (5.5% W)	2.6	0.25
A — ghost (adopted)	**90px (5.0% W)**	**2.0**	**0.06**

Ghost was adopted because it removes colour entirely. Colour is what makes a mark fight with artwork — it can clash with a palette, and it behaves unpredictably across the range of backgrounds in our library. A relief that only modulates lightness cannot clash with any palette by construction, which makes it the only variant that is safe to apply library-wide without art-directing each case.

Thumbnail RMS 0.06 means the seal is effectively absent at feed scale, while the full-resolution asset still carries a legible seal.

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4. Why the bottom-right corner

Placement was chosen from viewing behaviour, not taste.

• **Image saliency is centre-biased.** Fixations cluster toward the middle of a

frame, so all four corners are low-attention real estate. Corner placement is therefore cheap wherever we put it.

• **Reading order makes the corners unequal.** In left-to-right cultures the

top-left is the entry fixation — the most expensive corner. The Gutenberg diagram labels the bottom-right the *terminal area*: the eye arrives there after the subject and the message, so a mark placed there interrupts nothing.

• **Convention buys us habituation.** The bottom-right is where viewers have been

trained by decades of photo credits and watermarks to expect a small mark, so they skip it automatically.

• **Convention also buys us defensibility.** The bottom-left is arguably the very

weakest-attention corner, but putting a legal disclosure in an unconventional spot invites the accusation that we hid it. The bottom-right is the corner an auditor checks first — which is exactly what we want from a disclosure.

Net: lowest practical cost in attention, highest discoverability when someone comes looking. Corner is a single configuration value and can be changed house-wide, but it should not be changed per asset — consistency is part of the defence.

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5. What the seal is

A circular stamp built from four concentric elements, all defined as fractions of the seal diameter so the design is resolution-independent:

Element	Geometry	Purpose
Outer double rule	inset 1.2% and 6.2% of diameter	Reads as a seal at a glance
Radial guilloche ticks	60 ticks, radius 32.5%-42.5%	Security-print texture; hard to fake casually
Ring microtext	"AI-GENERATED IMAGE ·" repeated, radius 37.5%	The actual disclosure, in words
Centre monogram	"AI", cap height 23.5%	Recognisable at the smallest sizes

Applied specification (the house standard):

Parameter	Value
Placement	Bottom-right, margin 3.0% of the short side
Diameter	5.0% of the short side, clamped to 64-260px
Relief strength	5.0 L* peak, bipolar; 6.0 L* single-direction
Visibility floor	3.5 L* — never eased below this
Colour	None. a* and b* are not modified
Pixels touched	~0.1% of the frame
Minimum input	512px short side

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6. Models and technology

No generative model is involved in the seal. This matters: a diffusion model asked to "add a seal" produces a different seal every time, cannot be audited, and would repaint the artwork. The seal is deterministic code compositing.

Generative models are involved only *upstream* — they produce the imagery we are disclosing. That is precisely the fact the seal declares.

Component	Technology	Role
Seal geometry	`build_master.py`, Pillow vector drawing at 2048x2048	Draws the master mask once, ever
Typography	Inter 4.0 (SemiBold, Bold), font hashes recorded	Monogram and ring microtext
Colour model	CIELAB, D65 white point, computed in NumPy float64	Perceptual lightness control; leaves hue untouched
Resampling	LANCZOS, 4x supersampled relief build	Sub-pixel relief edges without aliasing
Metadata	ExifTool 12.57, XMP + EXIF, IPTC `digitalSourceType`	Machine-readable provenance
Runtime	Python 3.11, Pillow 12.3, NumPy 2.4	Application and verification

The pinned master mask is the source of truth. `seal_master.png` is a 2048x2048 8-bit greyscale mask of the seal linework:

```
SHA-256  6a6089230d3904f1c4357314102e47fe30b54c1ee386324530664d84c7f511e2
```

The applier never re-draws the seal; it scales this asset. This is what makes the mark identical across every image, machine, font version and library version — and the applier **refuses to run** if the master's hash does not match, so a silently edited seal cannot enter the pipeline.

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7. How the seal is applied

1. **Load and hash-check the master.** Mismatch aborts the run.
2. **Compute geometry** from the image's short side: diameter, margin, corner position. Integer, deterministic, no rounding drift.
3. **Build three relief maps** for that exact diameter, cached per run:

- `bipolar` — the mask offset up-left as a lit rim, offset down-right as a pressed shadow, plus a faint core so the linework has body. This is the carved look. - `press` — darkening only, for ground pinned near white. - `emboss` — lightening only, for ground pinned near black. Single-direction maps lose the light/dark edge pairing that makes hairlines visible, so their strokes are lifted with a compressive curve (gamma 0.55); the opposite-sign ringing lobe left by LANCZOS is clamped to zero.

4. **Convert the seal region to CIELAB** in float64.
5. **Choose polarity and mode once for the whole seal.** The test is how much of the intended relief actually lands: for each candidate, compare the requested ΔL* against what survives clamping to the 0-100 L* range, and count the fraction of ink pixels that lose more than half. If the bipolar bevel loses more than 2%, the seal switches to a single-direction mode. **This decision is never made per pixel** — a per-pixel flip rectifies the relief and destroys the carved shape, which was a real defect caught in review and fixed.
6. **Apply the relief to L* only.** a* and b* are passed through untouched, so hue and saturation are mathematically preserved.
7. **Pull the relief back where sRGB cannot represent it.** Shifting L* on an already-saturated colour can push it outside the gamut; the clamp that follows would then move a* and b* too, which is a hue shift. Instead the relief amplitude is halved, per pixel, until the result is representable — up to seven times. It only ever weakens the mark locally; it can never invert or recolour it. On a fixture with a blown specular highlight under the seal this cut the worst chroma error from Δab 5.24 to **0.468**, i.e. from a visible-in-principle shift to pure 8-bit rounding.
8. **Write only ink pixels.** Pixels inside the seal box that the relief does not act on keep their original bytes rather than picking up rounding from the Lab round-trip.
9. **Save without gratuitous loss.** PNG stays lossless; JPEG is re-encoded at quality 97 with 4:4:4 chroma; WebP is lossless; ICC profiles are carried through.
10. **Write metadata**, then **log a manifest line** with input and output SHA-256, mask hash, geometry, relief mode, applied ΔL*, pixel count and timestamp.

Metadata written

Field	Value
`XMP-iptcExt:DigitalSourceType`	`...digitalsourcetype/trainedAlgorithmicMedia`
`XMP-dc:Description`	Human-readable AI disclosure incl. Art. 50 reference
`XMP-photoshop:Credit`	`AI-generated (generative AI)`
`XMP-photoshop:Instructions`	House flag `GHOST-SEAL-V1`, mask hash prefix, UTC timestamp
`XMP-xmpRights:UsageTerms`	Publish only with the visible seal intact
`EXIF:ImageDescription` / `Software` / `UserComment`	Same disclosure, EXIF-side

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8. Guarantees, and the evidence for them

Measured on the reference campaign asset (1792x2400):

Guarantee	Evidence
Identical seal every time	Two runs on the same input produce **pixel-identical** output; seal scaled from a hash-pinned master
Artwork not compromised	`artwork_untouched_outside_seal: true` — bit-identical outside the 90px seal box
Hue and saturation preserved	Max chroma shift Δ(a*,b*) = **0.486** on the campaign asset and **0.468** on a fixture with a blown highlight under the seal — both pure 8-bit rounding, because out-of-gamut cases are pulled back rather than clamped
Relief lands as specified	Max ΔL* in seal = **4.98** against a 5.0 requested
Minimal footprint	**0.0999%** of frame pixels changed
Findable on audit	Correlation against the reference relief = **0.895**
Works on any background	Selftest passes on pure white, pure black, mid-grey, full-range gradient, pixel noise, and a real photograph
Never double-stamped	Re-running on a sealed file skips it, including after all metadata is stripped (detected by pixel correlation r = 0.895)
Survives JPEG	Seal still detected at r = 0.868 after a quality-97 re-encode

The selftest is the gate

`ghost_seal.py selftest` builds hostile cases and asserts, per case: determinism, relief strength above the floor, chroma stability, detectability, zero modification outside the seal box, and idempotency. It also asserts that a 400px image is **skipped**, not stamped. It must print `"PASS": true` before any production run, and again after any change to the script, the master, Pillow, or the machine.

Three defects were found by this harness and fixed before adoption: the per-pixel polarity flip that rectified the relief on extreme backgrounds; a clipping metric that mistook an already-white background

for an unachievable relief and needlessly weakened the seal; and out-of-gamut clamping that shifted hue by up to ∆ab 5.24 on a blown specular highlight.

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9. Operating rules

Stamp masters, derive everything else. Apply the seal to the master asset, then produce crops, resizes and thumbnails *from the sealed master*. Never stamp a derivative: pixel-level re-detection depends on the original geometry, and a shrunken seal is neither legible nor findable. Images with a short side below 512px are skipped by design for this reason.

One image

```
python3 ghost_seal.py apply IMG.png -o out/IMG_sealed.png
python3 ghost_seal.py verify out/IMG_sealed.png --source IMG.png
```

A library — mirrors the input tree, one manifest line per file, resumable because sealed files are skipped:

```
python3 ghost_seal.py batch ./library --out ./library_sealed --recursive --jobs 8
```

Audit a published file — no original required:

```
python3 ghost_seal.py verify suspect.jpg
```

The manifest is the audit trail. `manifest.jsonl` records, per asset: source and output SHA-256, mask hash, geometry, relief mode, applied ∆L*, pixels changed, whether metadata was written, and the UTC timestamp. Keep it in version control or ship it to a log store; it is what turns "we mark our AI images" into a claim we can evidence per file.

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10. Known limits — disclose these internally

- **Resized or re-cropped derivatives** keep their metadata but may fall below the pixel-detection threshold. Derive from sealed masters.
- **Extreme high-frequency texture** (uniform pixel noise) can hide a 5 L* relief from re-detection. The relief is still applied and the metadata still discloses.
- **Metadata is unsigned and strippable**, and many platforms strip it on upload. The visible seal is what survives a platform re-encode; C2PA signing is the answer if we need tamper evidence.
- **CMYK input is converted to RGB** and flagged in the manifest. Seal the RGB master; let prepress convert.
- **Legal scope is not settled here.** See section 2.

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11. Change control

The seal is a house standard, and its value comes from being the same everywhere.

- Changing **strength, size or corner** = editing `SPEC` in `ghost_seal.py`. Re-run the selftest, re-seal with `--force`.
- Changing the **seal artwork** = editing `build_master.py` geometry, rebuilding the master, re-pinning its SHA-256 in the skill metadata and in this document, then re-sealing the library with `--force`.
- Either change needs explicit sign-off, and the old and new mask hashes recorded.

Two different seals in circulation is worse than either seal alone: it invites the question of which assets were marked and which were missed.

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12. File inventory

File	SHA-256	Role
`seal_master.png`	`6a6089230d3904f1c4357314102e47fe30b54c1ee386324530664d84c7f511e2`	The pinned seal geometry. Source of truth
`ghost_seal.py`	`2415519056bad73984ea144a56a055a2c2f2ca01cbca031ecc07017566578e47`	Applier, verifier, batch runner, selftest
`build_master.py`	`0592891e326c9a3801a4a79ee6efea26d264185999fa8ac715ccdb8f0aa6f532`	Regenerates the master. Run only to change the standard
`seal_master.json`	—	Pinned mask hash, geometry table, font hashes
`verify_tree.py`	`3aa2a0c3514b94653d1992f03a0ab7335716afe79d8eba195f540088af15a0a4`	Publish gate: non-zero exit if anything unsealed ships
`tests/reference.png`	`a15f4f79cd40157d170b1a8cecef66031541f760ce27dac460337ffe2e0f5dc7`	Deterministic CI fixture (hostile ground under the seal)
`manifest.jsonl`	—	Append-only per-asset audit trail

Reference `verify` output on the campaign asset:

```
  seal_present_pixels ............ true
  seal_correlation ............... 0.8954
  geometry ....................... x 1648, y 2256, diameter 90 (5.02% of short side)
  mask_sha256 .................... 6a6089230d3904f1c4357314102e47fe30b54c1ee386324530664d84c7f511e2
  machine_readable_provenance .... true
  house_flag ..................... true
  artwork_untouched_outside_seal.. true
  max_chroma_shift_ab ............ 0.486
  max_dL_in_seal ................. 4.98
  pixels_changed_pct ............. 0.0999
  PASS ........................... true
```